Reg. No. | | | | | | | | | | |

[14EC35]

THIRD SEMESTER B.TECH EXAMINATION DECEMBER-2015

FUNDAMENTALS OF DIGITAL ELECTRONICS

Time:3 Hours **Maximum Marks:70**

Instructions: Answer any **FIVE** full questions. Selecting at least TWO from each part.

PART - A

1. a. What is a gray code? Explain the process of converting a number from gray code word to binary. **03**

   b. Implement the Boolean expression for Ex-OR gate using NAND gates. **05**

   c. Verify that the following operations are commutative but not associative. **06**

      i. NAND

      ii. NOR

2. a. Simplify the following 5 variable Boolean expression using Quine McCluskey method. **08**

   $F=\Sigma m(0,1,9,15,24,29,30)+d(8,11,31)$

   b. Design a logic circuit with 4 inputs P,Q.R,S that will produce a '1' output only whenever two adjacent input variables are 1s. P and Q are to be treated as adjacent in this problem. **06**

3. a. Design 5:32 decoder using 2:4 and four 3:8 decoder IC's. **05**

   b. Implement the given function in PAL **09**

   $A(x,y,z)= \Sigma m(1,2,4,6)$

   $B(x,y,z)= \Sigma m(0,1,6,7)$

   $C(x,y,z)= \Sigma m(2,6)$

   $D(x,y,z)= \Sigma m(1,2,3,5,7)$

4. a. Explain the following characteristics of digital ICs **05**

      i. Propagation delay

      ii. Power Dissipation

      iii. Fan-in

      iv. Fan-out

      v. Noise Margin

   b. What are the advantages of CMOS logic? Explain CMOS NAND with the help of a neat circuit diagram. **09**

PART-B

5. a. Distinguish between asynchronous and synchronous circuits. **04**

   b. Explain the operation of a switch debouncer circuit using SR latch. **06**

   c. Explain significance of clock and types of clock triggering. **04**

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6. a. List the types of shift registers. Discuss 4 bit Bidirectional shift register with a neat circuit. **07**

b. Design a counter with T flip flops that goes through the following binary repeated sequence:0, 1, 3, 7, 6, 4 **07**

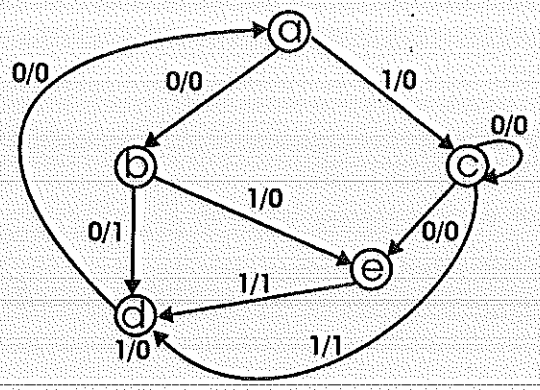
7. a. Differentiate between Moore model and Mealy model. **03**

b. Write a shot note on: **04**

i. Excitation and output expressions

ii. State reduction

c. Obtain the reduced state table and reduced state diagram for a sequential circuit whose state diagram for a sequential circuit whose state diagram is shown in figure below. **07**



8. a. Explain static RAM and Dynamic RAM. **04**

b. Design a $2^K \times n$ RAM has an Internal decoder of $K \times 2^K$ where n number of OR gates and each OR gates has 2^K inputs. **07**

c. Write a short note BT polar RAM cell. **03**